HISTORAG – Retrieval-Augmented Histopathology Atlas

TL;DR

This project investigates whether retrieval-augmented methods can be used to build an interactive histopathology atlas from whole slide images (WSIs). Image patches extracted from slides are embedded using pretrained vision or vision-language models and stored in a vector database. The system should allow image or text queries to retrieve visually similar tissue regions and evaluate retrieval performance on labeled histopathology data.

Background

Digital pathology enables large-scale analysis of whole slide images (WSIs) containing gigapixel-resolution tissue scans. Pathologists learn tissue morphology largely by studying many examples in atlases and case collections. Computationally, a similar paradigm can be implemented using content-based image retrieval (CBIR): given a query patch, the system retrieves visually similar tissue regions from a database. Recent advances in embedding foundation models allow images to be mapped into a shared semantic feature space. Combined with vector search systems, this enables efficient similarity retrieval over large datasets, enabling interactive exploration of pathology datasets.

Overall goal

Develop and evaluate an LLM-mediated, retrieval-based system for histopathology patches extracted from WSIs.

Minimal viable product (Phase 0 deliverable):

The MVP should implement a complete end-to-end pipeline that extracts image patches from whole slide images (WSIs), computes feature embeddings for each patch using a pretrained model, and stores these embeddings in a vector index for efficient similarity search. Given a query image patch, the system should retrieve the most similar patches from the database.

Tasks

- Data preparation (Extract image patches from WSIs using e.g. OpenSlide and store metadata (coordinates, slide ID, labels if available))
- Baseline embeddings using pretrained models (CLIP, CNNs, or vision transformers).
- Vector search using vector index (e.g., FAISS) and nearest-neighbor retrieval
- Evaluation (top-k accuracy, mean average precision, typical retrieval errors)
- Visualization (Create a simple interface showing query patches and retrieved results.
- Pro-1: enable text-based search using CLIP embeddings
- Pro-2: integrate a lightweight LLM to generate descriptions from retrieved patches
- Pro-3: extend retrieval to cross-slide similarity search